TRANSPORTATION / RIDE-HAILING

BANGALORE

Rapido Ride Analysis

Understanding Demand, Cancellations & Revenue in Bangalore

Team Members

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Mentors

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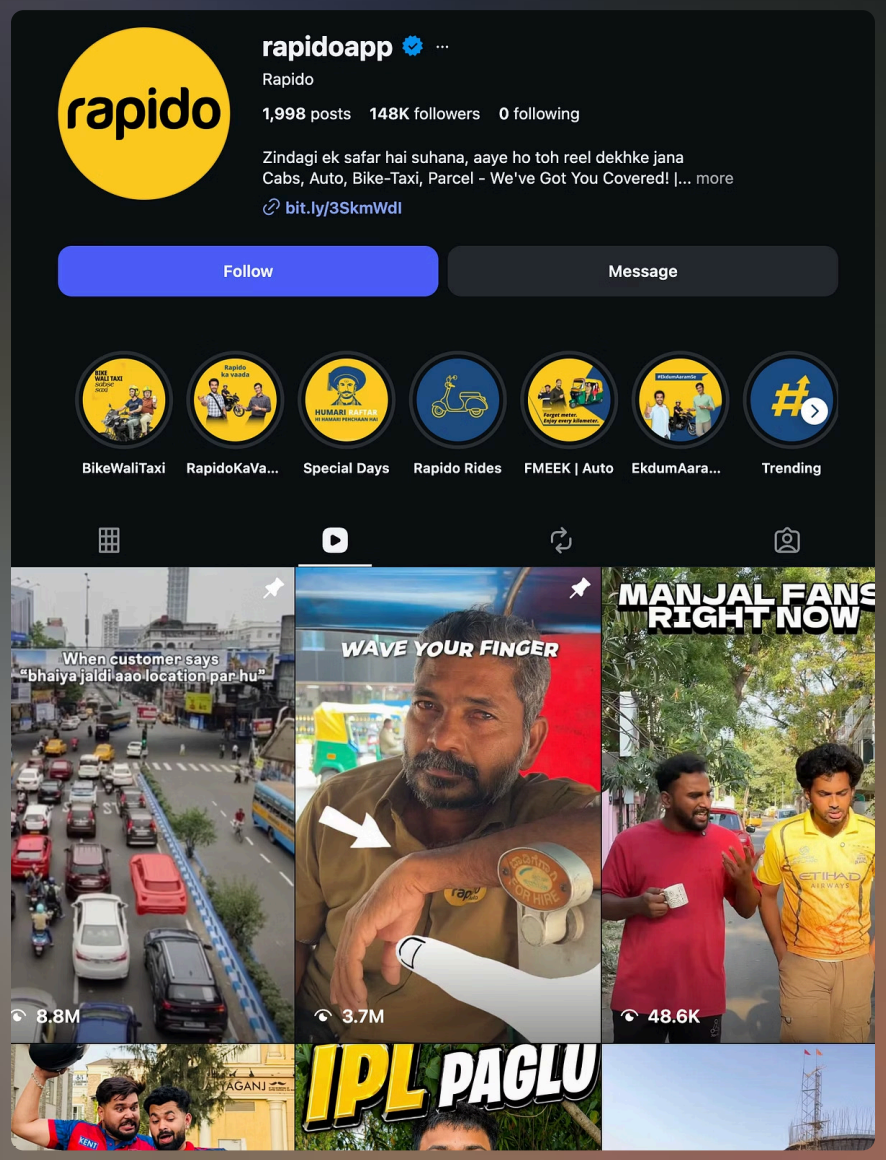
Project Details

- **Sector:** Transportation / Ride-Hailing
- **Institute:** Newton School of Technology, Rishihood University
- **Submission Date:** April 29, 2026



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EXECUTIVE SUMMARY

Executive Summary

Rapido Ride Analysis

Problem

Rapido faces a **10.4%** ride cancellation rate in Bangalore, leading to **₹4,000+** daily revenue loss. The issue is not low demand but supply failure in peak hours within high-value zones.

Approach

Analyzed **49,801** ride records using a Python ETL pipeline. Engineered **17** features including **demand_intensity** (z-score), **peak_hour**, and **fare_per_km**. Applied statistical testing and built a Tableau dashboard.

Key Insights

- High-demand zones have the highest cancellations: **Demand Paradox**
- Peak-hour cancellations rise **15–20%**
- Bike services maintain **92%+** completion rates
- Top zones generate **40%** of revenue but lack supply

Recommendation

A three-tier strategy — **incentives**, **rebalancing**, and **surge pricing** — can recover **₹1.44M annually** and reduce cancellations to **8%**.

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BANGALORE RIDE-HAILING | BUSINESS CONTEXT

Sector & Business Context

Urban ride-hailing in Bangalore faces a classic supply-demand mismatch: high demand, heavy congestion, and inconsistent driver availability.

Sector Overview

Urban ride-hailing in Bangalore operates under **heavy traffic congestion** and **fragmented driver supply**. Demand is high but unevenly serviced.

Stakeholder

Operations Manager responsible for fleet allocation, incentives, and service reliability.

Business Impact

- **High cancellations** reduce customer trust
- **Uneven supply** leads to revenue leakage
- **Lack of surge pricing** reduces driver motivation
- **Competitors gain** during peak failures

Core Insight

Solving **zone-level supply-demand imbalance** is the highest-ROI lever.

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ANALYSIS FRAMEWORK

Problem Statement & Objectives

Understand what drives cancellations, where revenue is being lost, and how to optimize driver allocation across Bangalore's ride-hailing network.

Problem Statement

How do service type, time, and location influence cancellations and revenue, and where should driver allocation be optimized?

01

Quantify demand vs cancellation relationship

02

Identify high-revenue zones with supply gaps

03

Build KPI framework for operational monitoring

Scope

- **49,801 rides** analyzed from April–August 2024
- Service types: **Bike, Auto, Cab, Parcel**
- Focus on **time-based** and **geographic** analysis

Success Criteria

Identify actionable zones for driver reallocation

Result: **5 high-demand zones** identified with supply gaps

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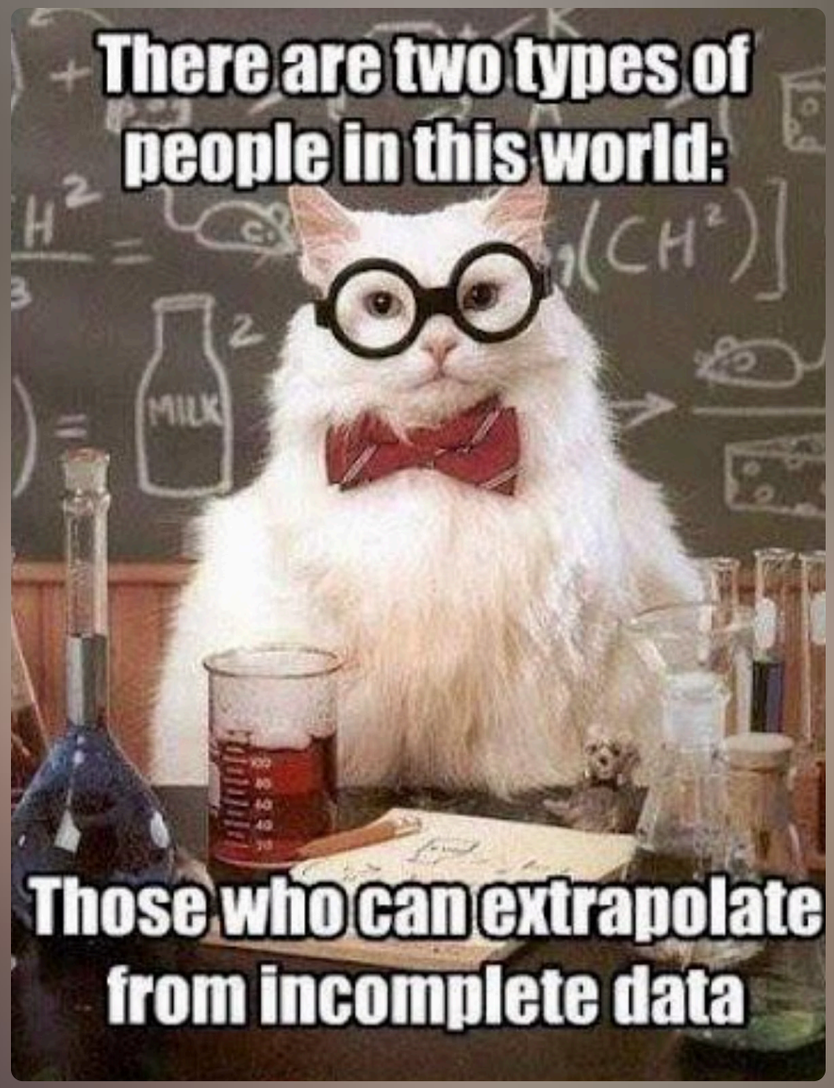

DATA ENGINEERING | PIPELINE

Data Engineering & Pipeline

From raw Kaggle data to a refined, analytics-ready dataset.

Data Source

Kaggle dataset

51,000 rows · 13 columns

Data Preparation

- Removed invalid distance/duration
- Deduplicated rides
- Standardized text fields

Data Cleaning

- Cancelled rides set to **₹0**
- Recalculated `total_fare`
- Normalized service names
- Parsed timestamps

Final Dataset

49,801 rows · 30 features

Zero nulls, analytics-ready

Outcome: Clean, consistent data foundation built for reliable analysis and reporting.

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DATASET PROFILE

Data Description

Dataset Overview

Source: Kaggle

Records: 49,801

Features: 30

Time: April–August 2024

Key Columns

- **ride_status** → Target variable
- **total_fare** → Revenue metric
- **services** → Segmentation
- **hour** → Time analysis
- **demand_intensity** → Demand metric
- **avg_speed_kmh** → Efficiency
- **fare_per_km** → Unit economics

Data Quality Fixes

1,199

Invalid rows removed

30

Features engineered

Missing fares were standardized and service naming inconsistencies were resolved.

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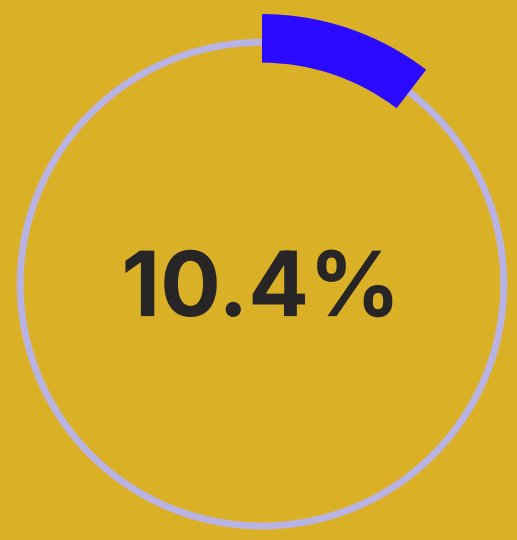
PERFORMANCE OVERVIEW

KPI Framework

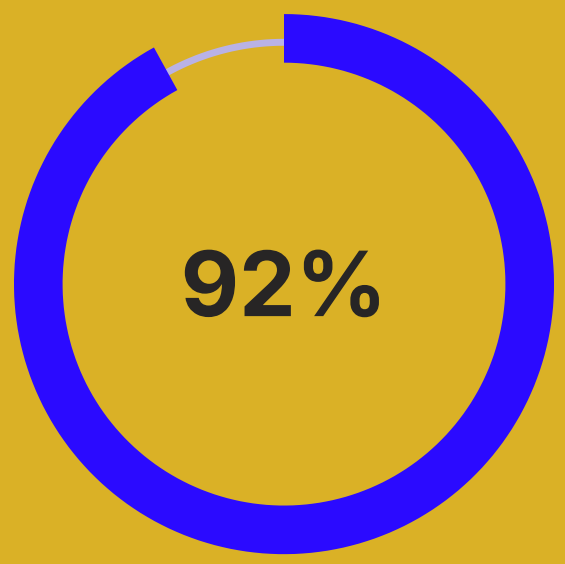
Core operational, revenue, and demand indicators tracking service health and monetization performance.

Service Quality



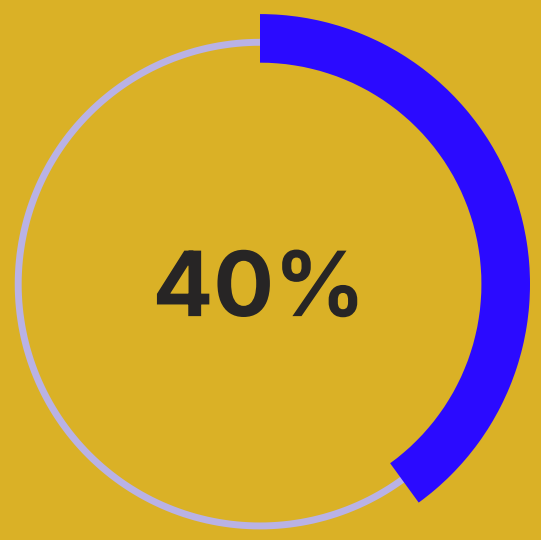
Ride Cancellation Rate

Key inefficiency metric



Bike Completion Rate

Service benchmark



Top Zone Revenue

Geographic concentration

Revenue Metrics

₹68.4L

Total Revenue

Baseline

₹153

Per Ride Average

Revenue per completed ride

₹20.94

Fare per KM

Pricing efficiency

Demand Metrics

17

Features Engineered

From raw data

15-20%

Peak Cancellation Increase

vs. off-peak

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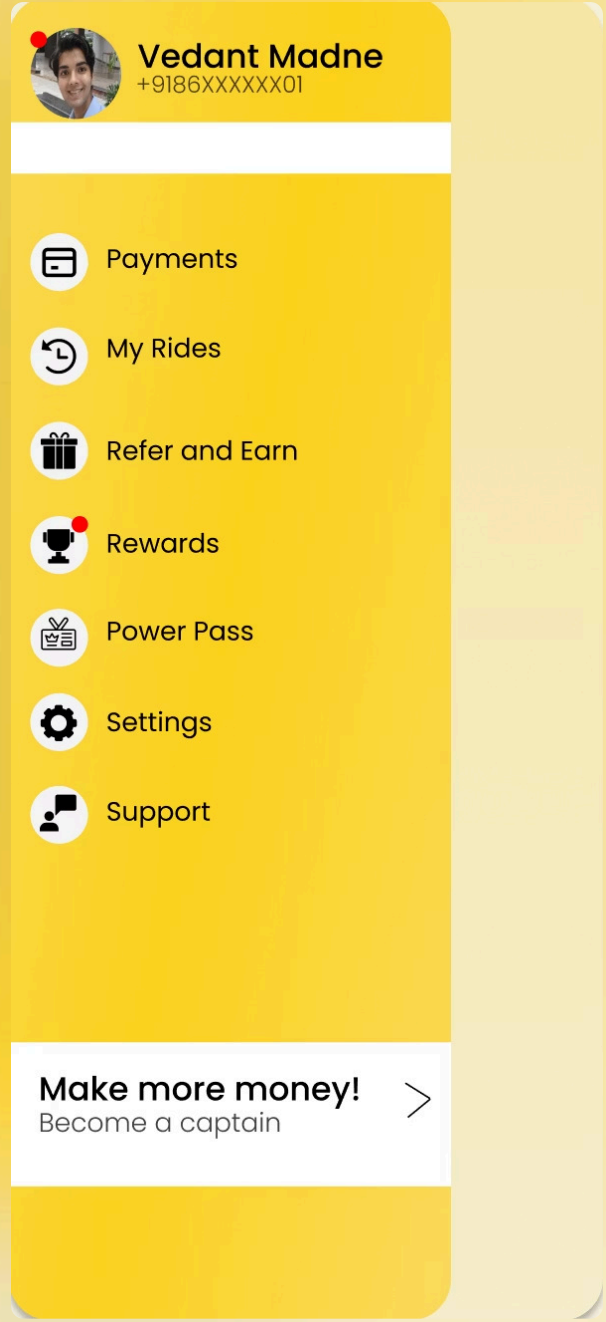

Exploratory Data Analysis

Service Distribution

Bike dominates volume while **Cab Economy** drives revenue. **Parcel** and **Auto** services show niche demand patterns.

Temporal Patterns

Peak cancellations occur **8–10 AM** and **5–7 PM**. **Weekends** show balanced demand with lower cancellation rates.

Distance & Pricing

Strong correlation **(r = 0.798)** between distance and fare. **No surge pricing** observed during peak demand.

Geographic Insights

Koramangala, **HSR Layout**, and **Indiranagar** generate **40%** revenue. Same zones show highest cancellations.

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STATISTICAL ANALYSIS

Statistical Analysis

Key statistical signals reveal where demand is concentrated, why completions drop, and which operational levers matter most.

Demand Paradox

High-demand zones paradoxically show lower completion rates due to insufficient driver supply during peak hours.

Demand Segmentation

Z-score identifies high-demand clusters in top zones. These zones require immediate supply optimization.

Hypothesis Testing

Chi-squared test results: **$p < 0.05$**
Cancellations are not random events but systematically linked to demand intensity.

Correlation Insights

- **Distance vs Fare:** strong positive correlation ($r = 0.798$)
- **Demand vs Cancellation:** negative correlation (higher demand → more cancellations)
- **Speed vs Completion:** strong positive impact

Tools Used

- Python
- Pandas
- SciPy
- Tableau

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Dashboard Walkthrough

Interactive views designed to support faster, more confident operational decisions - https://public.tableau.com/app/profile/aniket.pathak/viz/Rapido_Rides_Analysis/Overview

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Executive View

KPIs and revenue by service type with summary metrics for strategic decision-making.

Operational View

Heatmaps showing cancellation hotspots and supply-demand gaps across geographic zones.

Temporal View

Trends across peak versus off-peak hours with hourly patterns and day-of-week analysis.

Interactivity

Filters for service type, time period, and geographic zones enabling real-time analysis.

Outcome

Real-time decision support for the operations team to optimize driver allocation and reduce cancellations.

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Executive Summary

Key Insights

Critical patterns shaping demand, supply, revenue, and cancellations across the platform.

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1. Demand Paradox Exists

High-demand zones see lower completion rates because driver supply can't keep up.

2. Peak-Hour Revenue Loss

₹4,000+ in daily revenue is lost during peak hours from unfulfilled rides.

3. Two-Wheelers Lead Reliability

Bike services deliver the best reliability, with **92%+** completion rates.

4. Revenue Is Concentrated

Just **three zones** generate about **40%** of total revenue.

5. No Surge Pricing

The platform has **no dynamic surge pricing** to lift supply during peak demand.

6. Speed Drops Increase Cancellations

Lower ride speeds from congestion are strongly tied to higher cancellations.

7. Pricing Engine Works

The fare engine is performing correctly, pricing by **distance** and **service type**.

8. Cab Economy Struggles

Cab Economy underperforms most during high-demand periods.

9. Weekends Behave Differently

Weekend demand is more balanced, with **lower cancellations** than weekdays.

10. Cancellations Are Predictable

Cancellations are driven by **demand**, **hour**, and **location** — not randomness.

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RECOMMENDATIONS

Recommendations

Three focused actions to improve supply, reduce cancellations, and recover lost peak-hour revenue.

Geographic Incentives

Target top revenue zones with tailored driver incentives during peak hours to improve supply in critical areas.

Supply Rebalancing

Implement proactive driver fleet repositioning, particularly for bike services, before anticipated peak demand windows.

Dynamic Surge Pricing

Introduce dynamic surge pricing mechanisms in high-congestion and high-cancellation zones to balance demand and supply.

Expected Impact

Cancellation Reduction

8%

Anticipated reduction in overall ride cancellation rate from **10.4%** to **8%** within 6 months.

Annual Revenue Recovery

₹1.44M

Projected annual recovery of approximately **₹1.44 million** from previously unfulfilled peak-hour rides.

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IMPACT VALUE

Impact & Value

Key outcomes from the recommended improvements, highlighting financial upside, operational efficiency, and long-term strategic value.

Revenue Impact

₹1.44 Crore annual recovery potential from reduced cancellations and optimized operations.

Operational Gains

Anticipated **15%** reduction in customer wait times, leading to improved service efficiency.

Data Value

Development of a robust and reusable **ETL pipeline** and engineered features for future analyses.

Strategic Outcome

Enhanced customer retention, elevated service reliability, and strengthened brand loyalty.

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ANALYSIS SUMMARY

Limitations & Next Steps

Refined priorities to strengthen analytical depth, improve operational responsiveness, and scale impact.

Key Limitations & Constraints

Simulated Dataset

Analysis was based on a synthetically generated dataset, which may not fully capture all real-world complexities.

No Driver Behavior Data

Absence of granular driver data (e.g., driver acceptance rates, driving patterns) limited deeper behavioral insights.

No External Factors

External variables like weather, local events, or public transport disruptions were not included in the analysis.

No Real-time Integration

The current setup lacks real-time data ingestion and processing capabilities for immediate operational adjustments.

Limited Geographic Granularity

Analysis was restricted to predefined zones, potentially masking hyper-local demand-supply imbalances.

Strategic Next Steps

1. Predictive Modeling
Develop advanced models for forecasting demand and supply, including predictive cancellation likelihood.

2. Real-time Dashboard
Implement a dynamic dashboard with live data feeds to enable immediate operational interventions.

3. Multi-City Expansion
Extend the analytical framework and solutions to new operational cities for broader impact.

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